

Omar Younis

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SUMMARY

Software engineer with an **M.S. in Computer Science** and **seven prior years as a mechanical engineer**. Ships across the stack: Python data pipelines processing 1M+ records per minute at Qualcomm, CUDA GPU kernels, an iOS app on the App Store, and Coast Guard software adopted fleetwide. Comfortable learning any language the problem needs.

LANGUAGES AND TECHNICAL SKILLS

Languages: Python, C++, Swift, SQL, CUDA, VBA

Frameworks and Tools: PyTorch, TensorFlow, AWS (Lambda, Glue, S3, Athena, QuickSight), Google Cloud, BigQuery, Docker, Git, GitHub Actions, pytest, Vitest, MyPy, Ruff, Nsight Systems, SwiftUI, SwiftData, Streamlit, Claude Code

EDUCATION

M.S., Computer Science, California State University, Fullerton

May 2026

GPA: 3.86

M.S. Project: Parallelizing Brent's Method with CUDA

Relevant coursework: Data Structures, Algorithm Engineering, Advanced Algorithms, Operating Systems, Computer Communications, Compilers and Languages, Software Design and Architecture, Neural Networks and Deep Learning, Web Back-End Engineering, Mobile Development, Advanced Database Management

B.S., Mechanical Engineering, Worcester Polytechnic Institute (minor: Aerospace Engineering)

May 2016

EXPERIENCE

Software Engineer, United States Coast Guard Auxiliary, supporting USCG Sector San Diego, CA

Sep. 2022 - Present

- Shortened pilot training qualification tracking from 6 weeks to 2 days by building a reporting tool in **Python** and **VBA** that consolidates several data sources and the flight log database into a single dashboard. Adopted at every U.S. Coast Guard air station and still in daily use, with sole maintenance responsibility since 2022.
- Reduced helicopter parts search time by roughly 85% by developing a **Python** inventory system and database for the Sector San Diego aviation maintenance team.
- Built a maritime collision avoidance training system that computes closest point of approach and course/speed maneuver solutions from radar observations using vector-based relative motion algorithms, in **Python** and Streamlit with full pytest coverage and type-checked, linted code. Rebuilt it as a TypeScript monorepo to widen distribution beyond a single hosted app.

Software Engineer Intern, Qualcomm, San Diego, CA

May 2024 - Aug. 2024

- Reduced feature development and testing time by 40% by adding logging, monitoring, and alerting across the ETL pipeline stack.
- Built and industrialized AWS ETL pipelines processing 1M+ data points per minute for machine learning teams, using Lambda, Glue, S3, and Athena, in **Python** and **SQL**.
- Built QuickSight dashboards for stakeholders, including the top-level dashboard tracking every data flow in the system.
- Onboarded interns by writing an API development guide and configuring their development environments.

Teaching Associate, California State University, Fullerton

Jan. 2024 - May 2024

- Served as instructor of record for an undergraduate object-oriented programming course in **C++**, delivering all lectures and writing and grading all coursework for 64 students across multiple sections while carrying 12 graduate units.
- Wrote and taught an optional C++ and object-oriented programming primer, 12 hours across two weekend sessions delivered in person and online, for students entering without the introductory programming prerequisite or wanting a refresher; drew 5 to 10 attendees per session outside assigned course load.

Machine Learning Engineer, Elemeno AI, Remote

Aug. 2022 - Nov. 2022

- Improved delivery time prediction accuracy by 25% for Brazil's largest package delivery company by designing and training a feed-forward neural network in **PyTorch**.
- Ran exploratory analysis on large datasets with **SQL** and BigQuery, and built pipelines supporting client-specific model development and deployment on Google Cloud.

Lead Mechanical Engineer, R&D, D&K Engineering, San Diego, CA

Feb. 2019 - May 2021

- Secured a \$1M contract extension for a client in an at-risk trial period by leading a 20-person cross-functional team to develop next-generation ecoATM recycling kiosks.
- Raised in-house engineering efficiency by roughly 50% by building **Python** tooling for BOM cleaning, purchase order tracking, and data logging, later adapted for the manufacturing, electrical engineering, and management teams.
- Automated tracking of up to 5,000 parts per device into bills of materials, removing manual data entry from the mechanical engineering workflow.

- Developed embedded control software in **Python** for a Stanford University research prototype for 3D genome mapping, including sequential logic for sample preparation and the operator interface.
- Enabled Abbott Laboratories to monitor COVID-19 testing sites by building a **Python** logging system integrated with a camera array.

PROJECTS

radar-moboard, collision avoidance plotting trainer (**TypeScript**, Node, Vitest) 2026

- Ported a Python prototype to a four-package TypeScript monorepo in under two weeks while learning TypeScript, using Claude Code for debugging, refactoring, and generating language tutorials alongside each feature implemented.
- Enforced a branch protection rule requiring a passing CI suite via GitHub Actions before any merge to main, verifying geometry output against two independent answer keys across a 12-problem corpus of worked radar plots.

Parallelizing Brent's Method with CUDA, M.S. Project (**CUDA**, **C++**, **Python**) 2026

- Measured 35x kernel-level and 8.8x end-to-end speedup over a single-threaded C++ baseline by parallelizing 4M+ independent root-finding problems across GPU threads on an NVIDIA RTX 3080; a literature review identified no prior CUDA implementation of Brent's method. Profiled with Nsight Systems to isolate PCIe transfer cost.

Nahtadi, iOS app on the App Store (**Swift**, SwiftUI, SwiftData) 2025 - Present

- Shipped a privacy-first prayer times and Qibla app holding a 5.0 average across 8 ratings, computing results from astronomical algorithms with zero data collection and full on-device operation.
- Met WCAG 2.2 AA accessibility with VoiceOver and Dynamic Type support; backed by a separate zero-dependency **Python** library implementing Julian Day calculations, spherical trigonometry, and high-latitude adjustment across 8+ calculation methods.

On-Device LLM Summarizer (**Python**, **Swift**, PyTorch, MLX) 2025

- Gained +0.9 ROUGE-L over the base model by QLoRA fine-tuning Qwen2.5-1.5B on DialogSum, then quantized to 4-bit MLX at a measured 1.5 to 2 point cost and deployed it in a SwiftUI app running entirely on an iPhone 14 Pro A16 chip, below Apple Intelligence's hardware line.

Modular Compiler (**Python**) 2024

- Expanded a 339-line class project into a 4,000+ line modular compiler translating a Pascal-like language into executable Python, with a table-driven LL(1) predictive parser, semantic analysis, and AST-based code generation.

Cycloidal Drive Creator, open source, 28 GitHub stars (**Python**, Tkinter) 2023

- Generates cycloidal drive geometry and rotor equations with live preview before export; in active use by hobbyist and professional machinists.

LEADERSHIP AND AWARDS

- **Coast Guard Auxiliary Achievement Medal**, awarded by the Commandant of the U.S. Coast Guard, Mar. 2023, for superior performance of duty. Citation credits a training records program that spread fleetwide, increasing accuracy and reducing tracking time.
- U.S. Coast Guard Auxiliary volunteer since 2015; software developer for Sector San Diego since 2022.
- **Certifications**: Data Science Immersive, 500+ hours (General Assembly); Generative AI with Large Language Models (DeepLearning.AI and AWS); Algorithms Design and Analysis Part 1 (Stanford Online); AWS Serverless; Google Cloud Computing Foundations; Certified SolidWorks Associate.
- Fluent in several Arabic dialects; lived in three countries outside the United States.